

Introduction

This exploratory analysis investigates key business insights within the ‘onlineRetail2’ dataset. The dataset contains detailed transaction information including invoice numbers, timestamps, product prices, and purchase quantities. This collection of information provides a foundation for understanding customer purchasing behavior. Our goal is to uncover patterns and trends that can help to inform data-driven decisions for future inventory management and sales strategies. Our exploratory analysis begins with cleaning the data and reviewing the descriptive statistics to pinpoint any significant observations. We will later discuss the statistical tests that we performed based on these observations and relate the findings back to our business interests.

We selected this specific dataset because it is directly relevant to our group members’ personal interest in finance and retail analytics. Throughout this analysis, we will explore the patterns with the following questions in mind: Can we estimate the probability of future purchases by identifying associations between products? Are there meaningful insights that emerge from differences in product prices and quantities? By addressing these questions, we hope to extract actionable findings that can support informed business decisions.

Dataset Overview, Why We’re Interested

Taken from the UC Irvine Machine Learning Repository, our online retail dataset partition includes about 400K rows, for invoice dates from Dec 2010 to Dec 2011. The source data contains 541,910 rows. Each invoice item is one row. There are 135,080 rows with NA values, and 8905 rows with negative Quantity. After removing those rows, our dataset for analysis contains 397,925 rows.

With vast amounts of data now available, companies in almost every industry are focused on exploiting data for competitive advantage. Data mining is used for general customer relationship management to analyze customer behavior to manage attrition and maximize expected customer value. Probably the widest applications of data-mining techniques are in marketing for tasks such as targeted marketing, online advertising, and recommendations for cross-selling. Major retailers from Walmart to Amazon apply data mining throughout their businesses, from marketing to supply-chain management. For example, Target cares about consumers’ shopping habits, what drives them, and what can influence them. Consumers tend to have inertia in their habits.

Despite the large number of specific data mining algorithms developed over the years, there are only a handful of fundamentally different types of tasks these algorithms address. Co-occurrence grouping, also known as frequent itemset mining, association rule discovery, and market-basket analysis, attempts to find associations between entities based on transactions involving them. An example co-occurrence question: What items are commonly purchased together? Co-occurrence grouping considers similarity of objects based on their appearing together in transactions. Acting upon findings might require some creativity, but could suggest a special promotion, product display, or combination offer. Some recommendation systems also perform a type of affinity grouping by finding, for example, pairs of books that are purchased frequently by the same people. The result of co-occurrence grouping is a description of items that occur together. These descriptions usually include statistics on the frequency of the co-occurrence and an estimate of how surprising it is.

Our projected analysis includes association rules, analyzing supply and demand for complementary items, and unsupervised learning clustering items to group into categories. Clustering looks at similarity between objects based on the objects’ attributes. Caution for overfitting - looking too hard at the data may find patterns, but we seek to generalize to data not seen. We optimize leverage from our data, applying data mining to exploratory data analysis for a well-structured problem.

Exploratory Data Analysis, Descriptive Statistics

See presentation and slides for dataset preview, cleaning data, summary statistics, items sorted by quantity, total revenues by country, sample of items, duplicating item rows for quantity.

The goal of this exploratory analysis is to uncover patterns and trends. As usual of data analysis projects, we started preprocessing the data, including cleaning the date column so that it only displays the

date, trimming the spaces in the description column, and dropping the missing values along with the negative quantities that possibly refer to the returned items, which is a total of 8905 transactions including returned items. The exploratory data analysis continued with the most revenue-generating countries, being the United Kingdoms, Netherlands, and Eire respectively, and total revenues across time. This section concluded with descriptive statistics of the average number of each item sold, price and average items per invoice, respectively being 13.02, 3.12, and 26.45, as well as the highly sold items.

Techniques and Interpretation

Mining Association Rules

Although Quantity is a variable for every Invoice and Item, Quantity purchased doesn't affect Support, Confidence, and Lift. Confidence is the conditional probability of B given A.

$$Confidence(A \rightarrow B) = \frac{Support(A \cup B)}{Support(A)} = \frac{freq\{A, B\}/T}{freq\{A\}/T} = \frac{p(A \cap B)}{p(A)} = p(B | A)$$

Lift appears broadly across data science. Evaluate algorithms for targeting advertisements by computing the lift one gets for the targeted population. Calculate lift to help judge whether a repeated co-occurrence is interesting. Lift measures how much more frequently associations occur than we would expect by chance. Lift is calculated simply by applying basic notions of probability.

$$Lift(A, B) = \frac{p(A, B)}{p(A)p(B)}$$

The lift of co-occurrence of A and B is the probability of the itemset occurring in a transaction, compared to the probability of their occurrence if the items were unrelated to or independent of each other. An association is surprising if it contradicts something we already knew or believed. If associations from supermarket shopping cart data revealed that bread and milk are often bought together, we might not be surprised because many people buy milk and many people buy bread. So we would expect them to occur together frequently just by chance. We would be more surprised if we found associations that occur much more frequently than chance would dictate.

To prepare the dataset for association rules, we created the transaction baskets from invoice lines according to quantity as the original dataset is per item in an invoice, and then aggregated them per invoice, creating a total of 18,536 baskets. Then, the Apriori algorithm was run on the baskets to find the frequent itemsets with a minimum of 1% support. This implies itemsets that appear in at least 1% of the invoices are considered as frequent. This produced a total of 275 association rules. We have presented some examples of three- and two-item sets below. As an example from the two-item itemset consider the first line below that suggests an invoice that contains a pink plate will often contain the green one. In this regard, a confidence of 90.2% implies in 90.2% of the pink-plate invoices, green is also present, and support of 1.1% also suggests that these two appear in at least 1.1% of the invoices. Considering the baseline chance, lift of 61.9% indicates these two co-occur 61.9% more than if they were independent. Lastly, a conviction score of 10.034 suggests that a case of "pink plate without green plate" happens 10 times less often than if they were independent.

An alternative is to look at the difference of these quantities rather than their ratio. This measure is called leverage: $Leverage(A, B) = p(A, B) - p(A)p(B)$

Utilizing Quantity: Bins and Counts

In order to capture the purchase intensity that refers to "at what quantity, each item is related to the other item's quantity", we created five bins of equal quantities, and then attached the bin label to the item, and then re-ran the Apriori algorithm following rebuilding the invoice baskets. This will deliver the association rules between their purchase size and not just between their existence. This generated a total of 18 association rules. For example the very first one in the figures below suggests that in 70.2% of the very small orders of pink HBD bunting, there is also a very small order for the blue HBD bunting. In addition, they appear in at least 1% of the transactions, and they co-occur 47.48% more than if they

occurred by chance. Finally, the conviction score of 3.305 implies that “small purchases of pink without the small of green” occurs almost three times less often than chance.

Complementary Items

For the two items purchased together having the greatest lift. By invoice date and item, sum item quantities and get item’s average price.

In order to perform a deeper price-quantity analysis, we picked two items identified as complements through association rules, and for each date. Then, for each of them, we summed the total units of the item sold on that day across all the invoices and also calculated the average of the item price on that day across all the invoices. Later, we then merged these two series together by date. Finally, we plotted the green plate price and quantity versus each other as well as pink plate quantity versus green plate price. While the former indicates that the green plate quantity is almost always at the same low level with the increase in price, the second one suggests that with the increase in the price of the green plate, there is a tendency for the pink plate quantity to increase, which suggests these two are mostly substitutes to each other.

Item “REGENCY TEA PLATE GREEN” price and quantity have positive association, a rare effect. Since items are complementary, an increase in “REGENCY TEA PLATE GREEN” quantity corresponds to an increase in “REGENCY TEA PLATE PINK” quantity. If the price of a complementary item increases, demand for its complement usually decreases.

Clustering Items

Consider the task of identifying some natural groupings of business news stories released by a news aggregator. The objective of this example is to identify, informally, different groupings of news stories released about a particular company. This may be useful for a specific application; to get a quick understanding of the news about a company without having to read every news story; to categorize forthcoming news stories for a news prioritization process; or simply to understand the data before undertaking a more focused data mining project, such as relating business news stories to stock performance.

We finally performed K-mean clustering employing a total of 5 clusters, and assigned each item to the clusters. Each cluster grouped the items based on (0) kids craft/baking sets, (1) home decor, (2) general goods and decor, (3) retro or vintage items with some Christmas themed ones, (4) pink themed items. These clusters were generated based on word repetition. For example, If we look at cluster 3, we can observe that several items have the following words in their name/description: christmas, vintage.

Each item description is represented by a numeric feature vector using TFIDF scores scoring for each vocabulary word in the item description. TFIDF (Term Frequency times Inverse Document Frequency) scores represent the frequency of the word in the item description (document), penalized by the frequency of the word in the corpus (the raw data source of text). Words that tend to appear less frequently within the corpus have a TF-IDF higher score compared to those that are more common– from this, the corpus is used to create the matrix used to train the model (the numerical representation of our corpus). This is the reason why themes unique to each cluster can be observed. Each one created from reusing similar words, phrasings, or adjectives.

If we look at cluster 3, there is a linguistic pattern:

- “CHRISTMAS HANGING STAR WITH BELL”
- “DOG BOWL VINTAGE CREAM”
- “ROUND CAKE TIN VINTAGE RED”
- BISCUIT TIN VINTAGE RED”
- “CHRISTMAS CRAFT TREE TOP ANGEL”
- “AIRLINE BAG VINTAGE TOKYO 78”
- “LUCNH BAG VINTAGE DOILY”

We see that words such as “TIN”, “RED”, and “BAG” are commonly found in the cluster. However, we can infer that the words “VINTAGE” and “CHRISTMAS” have the highest TF-IDF scores and are acting as anchor signals influencing the centroid of that cluster. This inference can be made by comparing cluster 3 with a larger cluster like cluster 2. Cluster 2 contains some overlapping words with cluster 3:

Comparison to Cluster 2:

- “SUKI **SHOULDER** **BAG**”
- “**RED** **RETROSPOT** TRADITIONAL TEAPOT”
- “**RED** **RETROSPOT** BUTTER DISH”
- “**RED** FLORAL FELTCRAFT **SHOULDER** **BAG**”
- “**RED** **RETROSPOT** CHILDREN UMBRELLA”

Although “RED” and “BAG” appear frequently here and also appear in cluster 3, its TF-IDF score is lower due to its common appearance across the raw dataset. As a result, “RED” does not strongly influence the clustering compared to more unique terms like “VINTAGE”, “RETROSPOT”, or “CHRISTMAS”. This explains why clusters such as 3 and 4 form thematic groups while cluster 2 appears to cover a wider range.

We picked scikit-learn’s (sklearn.cluster) K-means for its ease of use. The algorithm uses Euclidean distance to calculate the similarity between centroids and how to group the data into clusters. The Euclidean distance uses our TF-IDF matrix to calculate which items are closest to each other, how the centroids are updated, and how the terms were grouped together.

The model uses Euclidean distance to calculate the similarity between centroids and how to group the data into clusters.

from the library scikit-learn (sklearn.cluster) to create the clusters. This information is important because it affects how Kmeans is directly creating the clusters.

Conclusion

Throughout this project, we followed an exploratory process to extract meaningful business insights from the ‘onlineRetail2’ dataset. We began with data cleaning and preparation, which included removing missing values & negative quantities, etc. to ensure that the dataset was reliable. With thousands of unique transactions, it was important to take out any potential outliers that could potentially skew the data incorrectly. This was followed by examining the descriptive statistics to get a big picture idea of how the data looked. This overview was the beginning step in choosing our statistical models to perform.

To address the first business questions, whether we can project probabilities of future purchases by identifying associations between products. We applied the Apriori algorithm to create itemsets and association rules. This revealed some strong patterns such as the pink and green plates frequently appearing together in transactions. By incorporating quantity bins, we were able to generate rules that captured not only the co-occurrence of the products, but also the size of order. Which gives deeper insight into how customers purchase complementary items at different quantities.

Regarding our second question, whether product prices and quantities revealed significant behavioral trends, we focused on the price column when trying to find relationships with item quantities. By aggregating the daily quantities sold and average daily prices, we were able to identify interesting dynamics between the Regency tea Plate Green and Pink items. There was a noticeable trend that as the price of the green plate increased, the quantity of pink plates also increased.

Now that we’ve identified some of the prominent relationships between items and prices, we can make business decisions with confidence. First, we would suggest prioritizing the coupling of the pink and green plates. Some marketing examples would be to create an enticing promotion that bundles the two items. This could be in the form of a discount, where customers may feel more inclined to purchase the two items since there's a monetary incentive on top of its natural appeal. Understanding the strong association between these two products could also be used for in-store or online advertising. Store managers could strategically position the items in certain aisles or towards the front of the store. Online

platforms could have the bundle shown on the home page of the website, offered with a special promotion. As a precaution, we would also suggest monitoring the shopping patterns closely as the marketing techniques are harnessed so that we can accurately manage inventory. This is an important factor in being able to supply the demand for these items. The analysis already showed that these items are frequently bought together, so we can imply that we will need comparable amounts of each item ordered during restock. All while considering the forecasted boost in sales due to our new marketing suggestions. Also, we want to keep in mind that our descriptive statistics did suggest certain peak times in purchases, hinting that there may be seasonality. This is also a big determinant in making ordering decisions as we do not want to disrupt cash flow and storage by overstocking during forecasted low periods. These tactics are the result of understanding what the data had to offer in terms of learning about the customer's shopping patterns. With proper implementation of these strategies, we expect to see a positive outcome of revenue specifically associated with the two items that we've identified.

Appendix

Clustering Technique: The goal is to present a description of the clusters, along with some items contained in that cluster. Convert collection of items to matrix of features. TF-IDF matrix is of dimension rows: number of items, and columns: number of terms. Increases in TF and increases in IDF cause increased TF-IDF values, for terms that occur frequently in an item's description, and occur rarely considering all items. We present the overall top 10 terms, and by cluster label, and samples of items by cluster assignment.

$$TF(Term, Item) = \frac{\text{Number Of Times Term Occurs In Item's Description}}{\text{Total Number Of Terms In Item's Description}}$$

$$IDF(Term, Item) = \frac{\text{Total Number of (NonUnique) Items In Corpus}}{\text{Number Of Item Descriptions Containing Term}}$$

The TF-IDF matrix contains rows as number of items, and columns as number of terms:

```
items = data["Description"]
vectorizer = TfidfVectorizer()
vec_items = vectorizer.fit_transform(items)
print(vec_items.shape)
```

✓ 2.2s

(397925, 2031)

Overall top 10 features: ['pink', 'polkadot', 'clock', 'alarm', 'bakelike', 'bag', 'blue', 'red', 'jumbo', 'green']

```

for k in range(0, 5):
    sample = results[results["cluster"] == k].sample(frac=0.0002)[["document", "cluster"]].to_string(index=False)
    print(sample)
    print("Items: ", len(results[results["cluster"] == k]))
    items = results[results["cluster"] == k][["document"]]
    vec_items = vectorizer.fit_transform(items)
    feature_names = vectorizer.get_feature_names_out()
    # Sum TF-IDF scores across all items for each term
    overall_tfidf_scores = vec_items.sum(axis=0).A.flatten()
    # Sort TF-IDF descending
    sorted_overall_indices = np.argsort(overall_tfidf_scores)[::-1]
    top_n_overall = 10
    #overall_top_features = [(feature_names[i], overall_tfidf_scores[i]) for i in sorted_overall_indices[:top_n_overall]]
    overall_top_features = [feature_names[i] for i in sorted_overall_indices[:top_n_overall]]
    print(f"Top {top_n_overall} Terms In Cluster Label {k}: {overall_top_features}")
    print("\n")

```

✓ 2.3s

document	cluster
SET OF 2 TRAYS HOME SWEET HOME	0
SET OF 4 ENGLISH ROSE COASTERS	0
MAGNETS PACK OF 4 SWALLOWS	0
SET/4 BADGES BEETLES	0
SET OF 3 WOODEN HEART DECORATIONS	0
PACK OF 72 SKULL CAKE CASES	0
SET OF 60 VINTAGE LEAF CAKE CASES	0
BAKING SET 9 PIECE RETROSPOT	0
WOODEN SCHOOL COLOURING SET	0
SET OF 10 LED DOLLY LIGHTS	0
SET/5 RED RETROSPOT LID GLASS BOWLS	0

Items: 53820
Top 10 Terms In Cluster Label 0: ['set', 'of', 'cake', 'pack', 'cases', 'retrospot', 'design', '12', 'tins', 'pantry']

document	cluster
VICTORIAN GLASS HANGING T-LIGHT	1
ANTIQUE SILVER T-LIGHT GLASS	1
SINGLE HEART ZINC T-LIGHT HOLDER	1

Items: 13810
Top 10 Terms In Cluster Label 1: ['holder', 'light', 'hanging', 'heart', 'glass', 'white', 'popcorn', 'silver', 'red', 'zinc']

document	cluster
OPEN CLOSED METAL SIGN	2
SPOTTY BUNTING	2
PANTRY ROLLING PIN	2
SPACEBOY BIRTHDAY CARD	2
SCOTTIES DESIGN WASHBAG	2
ELEPHANT, BIRTHDAY CARD	2
ENAMEL COLANDER CREAM	2
SPACEBOY ROCKET LOLLY MAKERS	2
LUNCH BAG APPLE DESIGN	2
FOUR HOOK WHITE LOVEBIRDS	2
IVORY KITCHEN SCALES	2
TUSCAN VILLA DOVECOTE	2
RED RETROSPOT CHARLOTTE BAG	2
SPACEBOY CHILDRENS CUP	2
ENAMEL BLUE RIM BISCUIT BIN	2
MAKE YOUR OWN PLAYTIME CARD KIT	2
JINGLE BELL HEART ANTIQUE SILVER	2
STRAWBERRY HONEYCOMB GARLAND	2
GARDENERS KNEELING PAD CUP OF TEA	2
TRAVEL CARD WALLET TRANSPORT	2
BLUE GLASS GEMS IN BAG	2
OWL DOORSTOP	2
SMALL HANGING GLASS+ZINC LANTERN	2
TEA BAG PLATE RED RETROSPOT	2
5 HOOK HANGER MAGIC TOADSTOOL	2
SLATE TILE NATURAL HANGING	2

```
LUNCH BAG RED RETROSPOT      2
CUPID DESIGN SCENTED CANDLES  2
HEART DECORATION PAINTED ZINC  2
    LUNCH BAG WOODLAND        2
    LUNCH BAG SPACEBOY DESIGN  2
        BABY BOOM RIBBONS      2
LADIES & GENTLEMEN METAL SIGN  2
WHITE SKULL HOT WATER BOTTLE  2
    PARLOUR CERAMIC WALL HOOK  2
        TOTE BAG I LOVE LONDON  2
            ORANGE TV TRAY TABLE  2
                BLUE HARMONICA IN BOX  2
DINOSAUR KEYRINGS ASSORTED    2
ENAMEL MEASURING JUG CREAM     2
MAGIC DRAWING SLATE LEAP FROG  2
DRAWER KNOB CRACKLE GLAZE IVORY  2
HEADS AND TAILS SPORTING FUN   2
    3 STRIPEY MICE FELTCRAFT    2
        YELLOW METAL CHICKEN HEART  2
FELTCRAFT PRINCESS LOLA DOLL   2
    HENRIETTA HEN MUG          2
        PARTY BUNTING          2
GARDENERS KNEELING PAD KEEP CALM  2
    JUMBO BAG APPLES            2
        SMALL RED BABUSHKA NOTEBOOK  2
            WRAP RED DOILEY        2
3 HEARTS HANGING DECORATION RUSTIC  2
Items: 266512
Top 10 Terms In Cluster Label 2: ['bag', 'red', 'retrospot', 'heart', 'design', 'lunch', 'box', 'metal', 'jumbo', 'white']
```

```
document cluster
TRAVEL CARD WALLET VINTAGE TICKET      3
VINTAGE DOILY TRAVEL SEWING KIT        3
    BISCUIT TIN VINTAGE RED             3
        GREEN VINTAGE SPOT BEAKER        3
SET OF 20 VINTAGE CHRISTMAS NAPKINS     3
    VINTAGE CHRISTMAS TABLECLOTH       3
SET OF 6 RIBBONS VINTAGE CHRISTMAS      3
Items: 36507
Top 10 Terms In Cluster Label 3: ['vintage', 'christmas', 'bag', 'jumbo', '50', 'doily', 'red', 'paper', 'leaf', 'kit']

document cluster
    TOY TIDY PINK POLKADOT              4
        MINT DINER WALL CLOCK            4
            SMALL CHOCOLATES PINK BOWL    4
FELTCRAFT HAIRBAND PINK AND PURPLE      4
    IVORY DINER WALL CLOCK              4
Items: 27276
Top 10 Terms In Cluster Label 4: ['pink', 'polkadot', 'clock', 'alarm', 'bakelike', 'bag', 'blue', 'red', 'jumbo', 'green']
```

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<https://archive.ics.uci.edu/dataset/502/online+retail+ii>

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